



Chapter 1 — Substrate

What a language model is, and what one run of it computes

The answer arrives a piece at a time

animation

a question typed, and the reply appearing left to right at natural speed

1600 x 422 px

The text you send is the **prompt**. Every character counts, spaces and typos included, and it can carry images too. The reply comes back a piece at a time.

observed

Substrate

Formation

Control

Measure

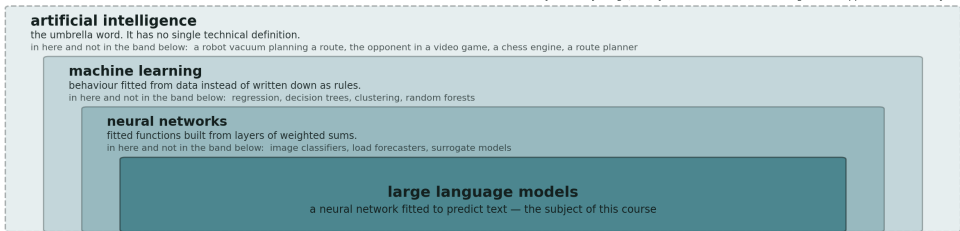
Failure

Limits

Language models are one branch of AI

Every band contains the one inside it. Not all artificial intelligence is a language model.

dashed = a boundary drawn by usage, not by definition — "artificial intelligence" is applied inconsistently



A **neural network** is a fitted function of layered weighted sums. A **large language model** is one fitted to predict text.

definition

Substrate

Formation

Control

Measure

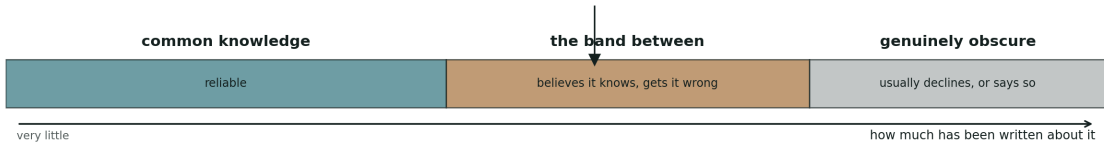
Failure

Limits

Reliability follows how much was written

Reliability tracks how much has been written, not how important the topic is.

roughly where your own research
literature sits



The **corpus** is the text training ran over. A prediction can be fluent and still wrong; what governs which is how much was written. **TODO(verify)**

definition



A model is numbers plus a program

the parameters
one long list of numbers

-0.0412	0.9037	0.1188	-1.2740
0.0006	0.4451	-0.3319	0.7702
0.0925	-0.0187	1.1046	-0.6538

... and so on, counted in billions

the program
the arithmetic that uses them

1. read the numbers from the list
2. multiply and add them against the input, layer after layer
3. hand back one score per token

Both are files on a disk. Together they are the model.

Parameters

also **weights**; *fitted*, not designed

Network

layered weighted sums

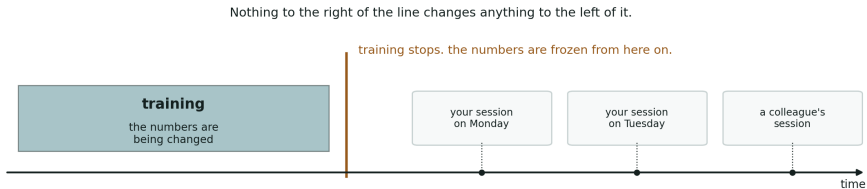
Size

how many were fitted

definition



Training set those numbers and then stopped



Training

the phase in which the numbers were changed

Inference

running the finished numbers on your text

Knowledge cutoff

the date training stopped

Session

one continuous conversation

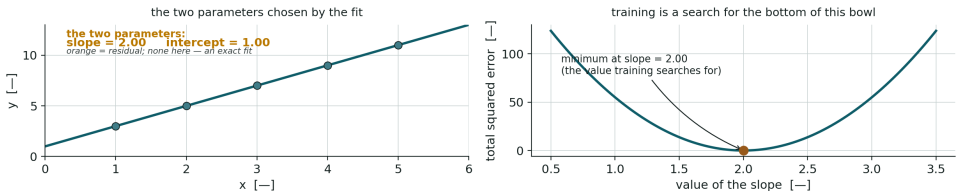
definition



Limits

Training searches for the smallest total error

$y = 2x + 1$ exactly — chosen so slope 2.00 and intercept 1.00 can be verified by hand. A real fit does not land on zero residual.

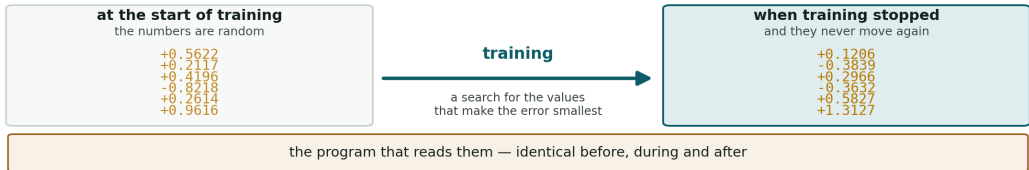


Two parameters here: **slope 2.00** and **intercept 1.00**. A language model runs the same kind of search over billions of them, with text as the data. There is no closed form, and it stops at a good enough set rather than the best possible.

computed

Training changed the numbers, nothing else

ILLUSTRATIVE values, drawn from a fixed seed. The movement is the point, not the numbers.



Neural network

layers of multiply-and-add,
named after a biological analogy

Learning

the search you just watched

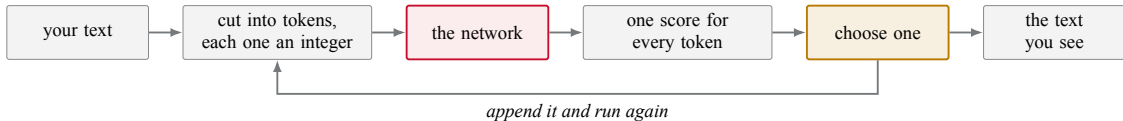
The brain comparison

correlational; the appendix
carries what it establishes

[V] Schrimpf, M., Blank, I. A., Tuckute, G., et al., 2021. *PNAS* 118(45):e2105646118, DOI 10.1073/pnas.2105646118.

[V] Hadidi, N., Fegghi, E., Song, B. H., Blank, I. A. & Kao, J. C., 2026. *Nature Communications* 17(1):5769, DOI 10.1038/s41467-026-72253-7.

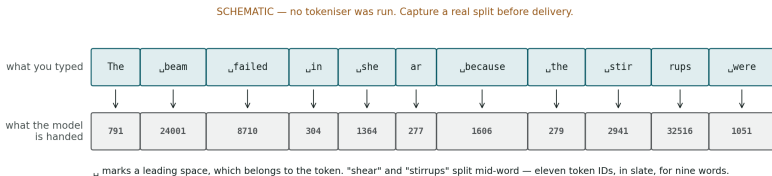
Text becomes numbers, then scores, then text



We walk this left to right. Some boxes take more than one slide. **Token** is a chunk of characters from a fixed list, defined next.

definition

Tokens come from a fixed list



* Not only text. An image or a PDF page is cut into small squares and turned into numbers in the same window.

The vocabulary

the model's list, not English

Token IDs

labels, not quantities

No letters

the network is handed integers

shear may split as *she*+*ar*: common
letter runs, not words

schematic

Substrate

Formation

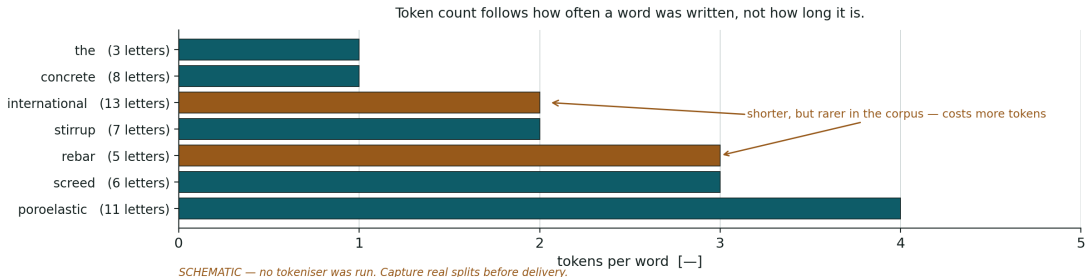
Control

Measure

Failure

Limits

Token count follows frequency, not length



Billed
per token

Context limit
counted in tokens

Speed
quoted per token

schematic

Substrate

Formation

Control

Measure

Failure

Limits

Training minimises surprise at the next token

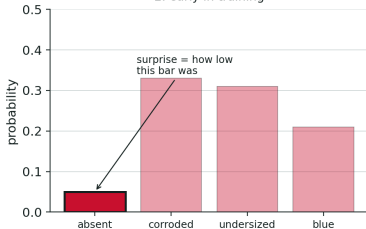
1. a training example

The beam failed in shear because

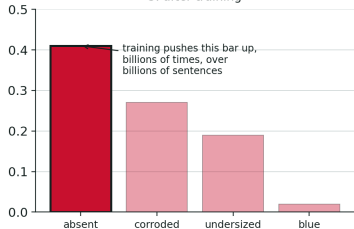
the stirrups were L

during training, the true
next token is known

2. early in training



3. after training



You cannot define "surprised" without a probability. The distribution is what training shaped.

Surprise is $-\log p$ for the token that actually came next, so the thing training shaped is a distribution over every **token** that could come next.

derived

Substrate

Formation

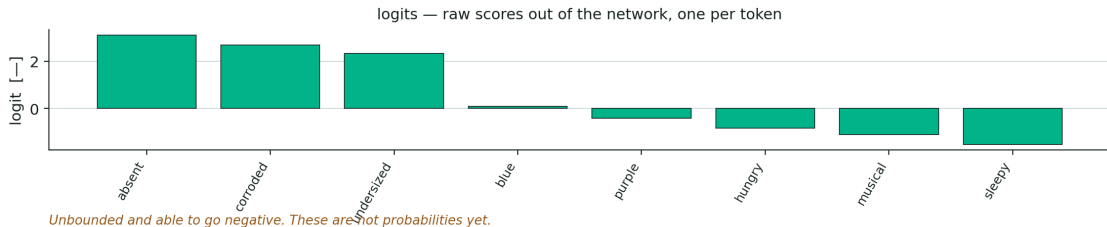
Control

Measure

Failure

Limits

The network returns one score per token



A **logit** is a raw score for one token, unbounded and able to go negative. One run of the network over the window is one **pass**. These are not probabilities yet, and the remainder.

definition

Substrate

Formation

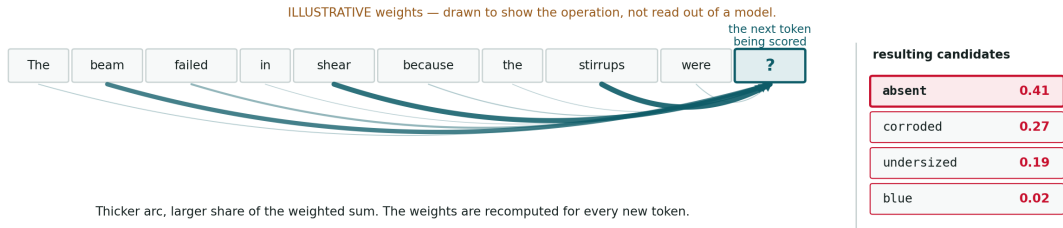
Control

Measure

Failure

Limits

Attention decides which earlier tokens matter



animation A3 replaces this figure — 1600 × 600 px

A **weighted sum**: the weight on each earlier position comes from comparing it against the position being scored. A **transformer** is the arrangement of that arithmetic.

[V] Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, L. & Polosukhin, I., 2017. *Attention Is All You Need*, arXiv:1706.03762.

Softmax turns scores into probabilities

$$p_i = \frac{\exp\left(\frac{\text{score for token } i}{\text{temperature}}\right)}{\underbrace{\sum_j \exp(z_j/T)}_{\text{sum over every token, so the result adds to 1}}}$$

As $T \rightarrow 0$ the top score wins outright. $T = 0$ **means take the maximum**, not divide by zero.

Worked case, two candidates only, so the sum checks by hand.

$$z_{\text{absent}} = \ln 0.41 \quad z_{\text{corroded}} = \ln 0.27$$

$$p = \frac{0.41}{0.41 + 0.27} = 0.603$$

Two in the denominator gives **0.603**. Over the whole vocabulary it is **0.41**, which is the number the figures carry. All quantities dimensionless.

computed

Always taking the top token repeats

animation

temperature zero, run long enough that the output falls into a loop

1600 x 422 px

Taking the top token every time is **greedy decoding**. It is observed to fall into loops: once a phrase appears, it raises the probability of its own continuation. **TODO(cite)**

observed

Substrate

Formation

Control

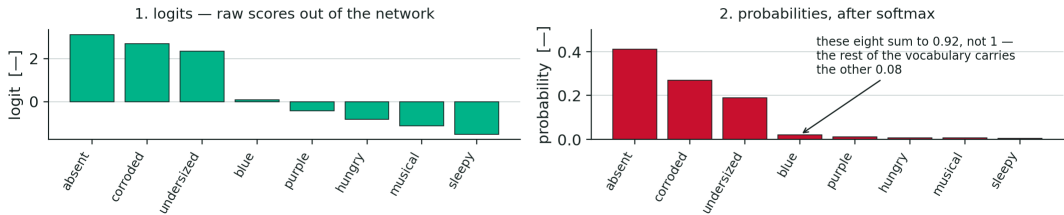
Measure

Failure

Limits

The token is drawn, not taken

illustrative logits — the softmax transform applied to them is exact, over these eight plus one term for the rest of the vocabulary



animation A4 replaces this figure — 1600×600 px — the bars must be identical on every draw

The bars are computed from the frozen numbers and never change. The **draw** does.

derived

Substrate

Formation

Control

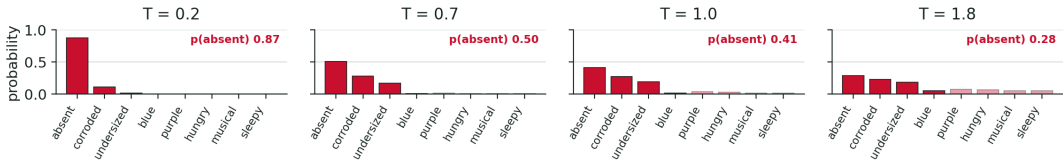
Measure

Failure

Limits

Temperature sets how sharply the top score wins

Same eight candidates, four temperatures — logits are $\log(\text{table probability})$, so softmax at $T = 1$ reproduces the table exactly



animation A5 replaces this figure — 1600×450 px

Low T

one token takes almost
everything

$T = 1$

the scores used as they are

High T

the **probability** spreads

computed

Substrate

Formation

Control

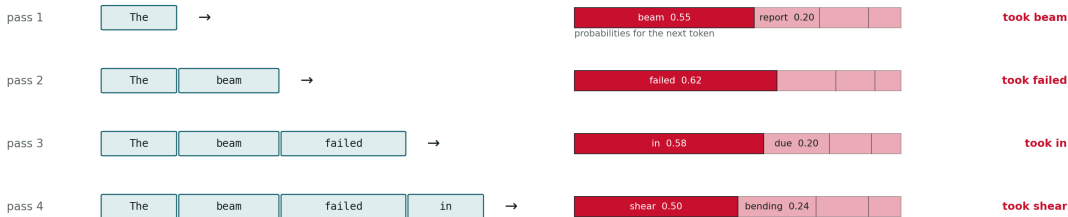
Measure

Failure

Limits

Each token is appended and everything reruns

ILLUSTRATIVE — the same frozen numbers are read on every pass. Only the text handed to them has grown.



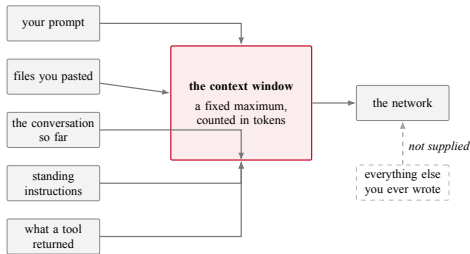
animation A7 replaces this figure — 1600 × 450 px

The **parameters** are unchanged on every one of them.

schematic



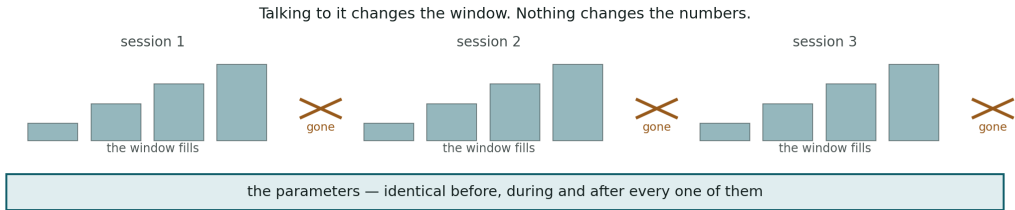
Everything given sits in the context window



Re-read in full every time the network runs. **Standing instructions** are placed in the window at the start of every session. A **tool** is a program run alongside the network, and its result is written into the window as tokens.

[V] Anthropic, 2026. *Code execution tool*, platform.claude.com — vendor documentation, fetched 2026-08-20.

The session ends and the window goes



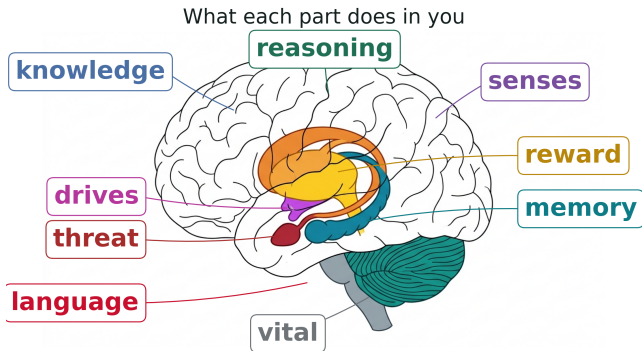
The network keeps nothing
nothing you sent changed it

Unchanged
the **parameters**, in every session

definition



Your brain divides the work into parts



schematic

Substrate

Formation

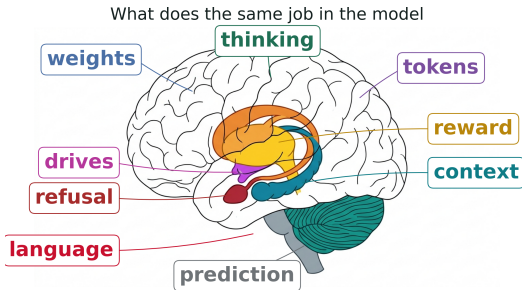
Control

Measure

Failure

Limits

One set of weights does every job



Language and thought are separable systems in people, a double dissociation, and one set of weights in the model.

[V] Fedorenko, E., Piantadosi, S. T. & Gibson, E. A. F., 2024. *Nature* 630(8017), DOI 10.1038/s41586-024-07522-w.

What we covered, and what is still open

What we covered

- ▶ A model is a list of **numbers** and a program that reads them
- ▶ Training fixed those numbers and stopped
- ▶ What it minimised was surprise, so the output is a **distribution**
- ▶ Text is cut into **tokens** from a fixed list
- ▶ Every pass scores every token, then draws one
- ▶ It sees only the context window, and discards it

What is still open

- ▶ Nothing in scoring text makes a thing that answers a question, follows an instruction, or declines a request. What was added?
- ▶ Why does it agree with you so readily?
- ▶ If it is reliable where much has been written, how do you tell which regime you are in?
- ▶ How would you ever know a change to your prompt actually helped?

derived